Network

Consensus layer

Copy page

Arc's Malachite consensus layer orders, validates, and finalizes blocks using a Tendermint-based Proof-of-Authority model.

Arc's consensus protocol provides deterministic finality in under one second. It is built on Malachite, a high-performance implementation of the Tendermint Byzantine Fault Tolerant (BFT) protocol, and uses a Proof-of-Authority (PoA) validator set.

As a developer, you don't interact with consensus directly, but it defines the guarantees you can rely on when building payment, trading, or settlement applications.

Core properties

Arc consensus is designed for institutional-grade performance and trust:

- **Deterministic finality:** A transaction is either unconfirmed or final. Once finalized, it cannot be reversed or reorganized.
- **Low latency:** Blocks finalize in less than one second under normal conditions.
- **High throughput:** Benchmarks show 3,000+ TPS with 20 validators and sub-second latency. Smaller validator sets can reach 10,000+ TPS.
- **Validator accountability:** Validators are regulated institutions with operational and compliance obligations.
- **Optimistic responsiveness:** The Tendermint protocol implemented by Malachite ensures block production and transaction confirmation proceeds as fast as the network permits, with no extra timeouts or artificial delays.

Proof-of-authority validator set

Arc uses a **permissioned Proof-of-Authority (PoA)** model.

- **Validators** are selected, known institutions with reputations, compliance requirements, and operational guarantees (such as uptime SLAs and SOC 2 certification).
- **Geographic distribution** ensures resilience. Validators run across multiple global regions.
- **Block production** is rotated among validators to ensure fairness and liveness.

This design provides stronger assurances for regulated finance by replacing anonymous economic incentives with institutional accountability.

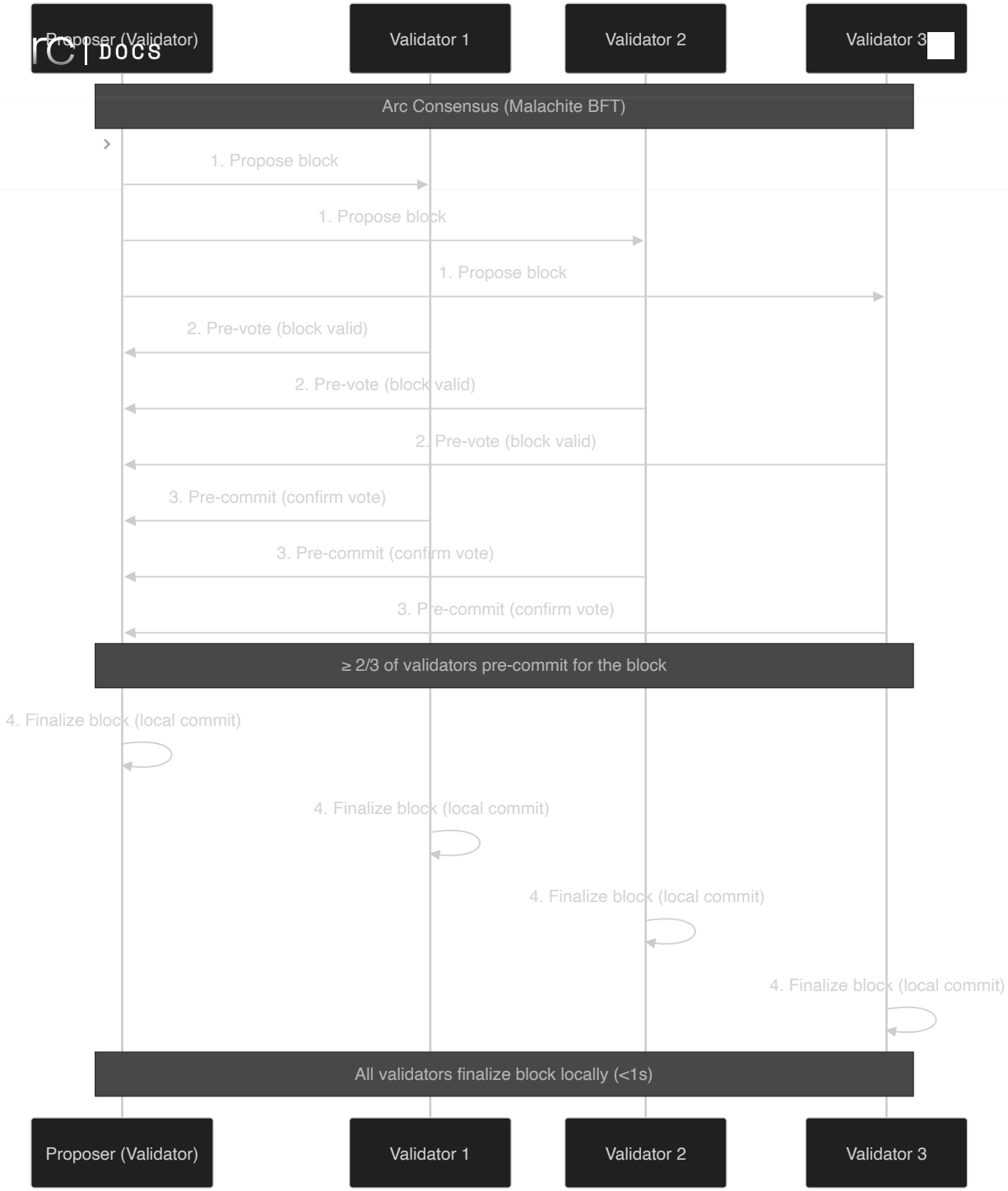
How Tendermint consensus works in Malachite

To order and finalize transactions, Arc uses the Tendermint BFT consensus protocol, implemented in the Malachite consensus layer.

At a higher level, Tendermint works as follows:

1. **Propose**
   - One validator is chosen as proposer for a round.
   - The proposer bundles transactions into a block and broadcasts it.
2. **Pre-vote**
   - Validators broadcast votes indicating whether they consider the block valid.
3. **Pre-commit**
   - Validators broadcast a second round of votes.
   - If more than two-thirds of validators pre-commit to the same block, that block gets committed.
4. **Commit**
   - The block is finalized and appended to the chain.
   - Transactions inside the block are now irreversible.

This two-phase voting process ensures consensus safety: two conflicting blocks cannot both be finalized. As a result, block reorganizations are impossible, and each block is finalized quickly and deterministically. The diagram below illustrates the high-level concept behind this process.



Deterministic finality

Unlike probabilistic models (like proof-of-work), Arc provides certainty about finality.

- Once a block is finalized, it cannot be reverted without collusion of at least two-thirds of validators.
- There is no need for developers to wait for multiple confirmations.
- Applications can release funds or complete trades immediately after confirmation.

For developers, this reduces complexity since you don't need rollback logic, and you can provide users with instant settlement guarantees.

›

Performance characteristics

Arc is engineered for low latency and high throughput. In testnet environments, you can expect performance characteristics similar to the following:

- **3,000 TPS** with 20 globally distributed validators.
- **<350 ms** finality under benchmark conditions.
- **>10,000 TPS** with reduced validator counts (for example, 4 validators).
- **Future roadmap** includes multi-proposer support (see below), which can increase throughput by ~10X, and consensus optimizations that can cut latency by ~30%.

These metrics make Arc suitable for high-frequency payments, trading, and settlement systems.

Multi-proposer

The Malachite roadmap includes a planned upgrade called multi-proposer. This feature allows multiple validators in the network to propose blocks in parallel, rather than sequentially. By enabling concurrent block proposals, multi-proposer can significantly increase network throughput and improve overall scalability.

Security guarantees

Arc combines protocol-level safety with institutional safeguards:

- **Safety:** With <1/3 faulty validators, consensus guarantees no conflicting blocks can be finalized.
- **Liveness:** The system continues to make progress as long as ≥2/3 of validators are online and honest.
- **Accountability:** Validators are regulated institutions with compliance obligations, making malicious behavior costly in the real world.
- **Resilience:** Geographic distribution reduces correlated downtime and attack risk.

Developer implications

For developers, consensus guarantees that:

- Your transactions settle instantly and irreversibly.
- You don't need to design around chain reorganizations or probabilistic confirmations.
- Arc can handle institutional workloads with high TPS and low latency.
- Validator accountability ensures the network remains secure and compliant, even at scale.

Roadmap

Malachite continues to evolve:

- **Multi-proposer support:** Multiple proposers per height increase overall throughput.
- **Latency optimizations:** New protocol variant reduces consensus rounds from three to two.
- **Permissioned Proof-of-Stake transition:** Over time, Arc may evolve from PoA to a permissioned PoS model, allowing broader validator participation while maintaining compliance.

These upgrades will further strengthen Arc as infrastructure for global financial applications.

Was this page helpful? Yes No

Contract addresses Execution layer

Links

Network status

Blog

Community

Arc homepage

Resources

Terms of service

Privacy policy

Cookie policy

Powered by mintlify

By using the Arc Network Testnet, you agree to the Terms of Service and Privacy Policy.